18/8/26

Quicksort:

Worst Case: $T(n) = T(n-1) + T(1) + \Theta(n)$

$\in O(n^2)$

Best Case:

$T(n) = 2T(n/2) + \Theta(n) \in O(n \log n)$.

10 : 90 Partitioning.

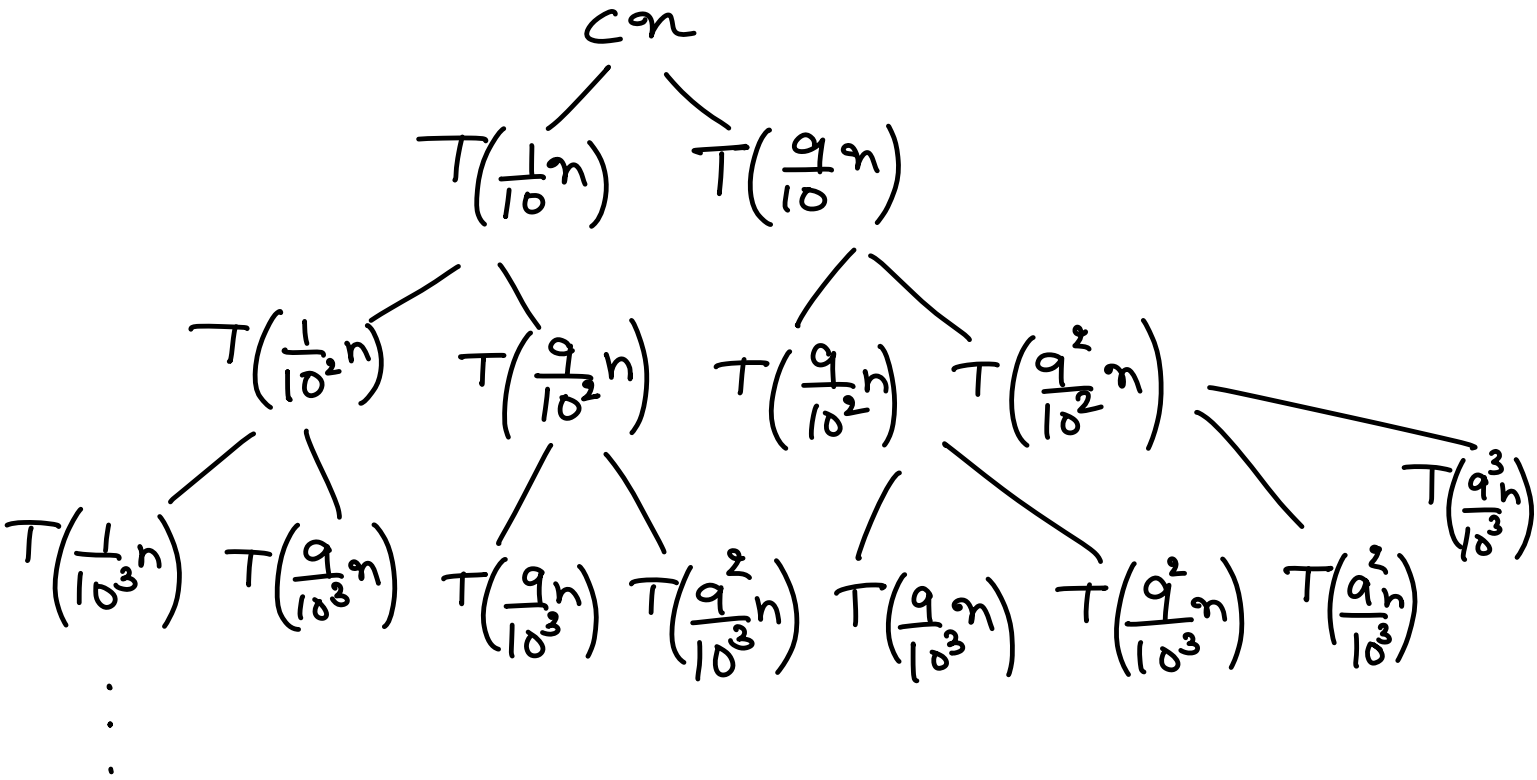
$$T(n) = T\left(\frac{1}{10}n\right) + T\left(\frac{9}{10}n\right) + \Theta(n)$$

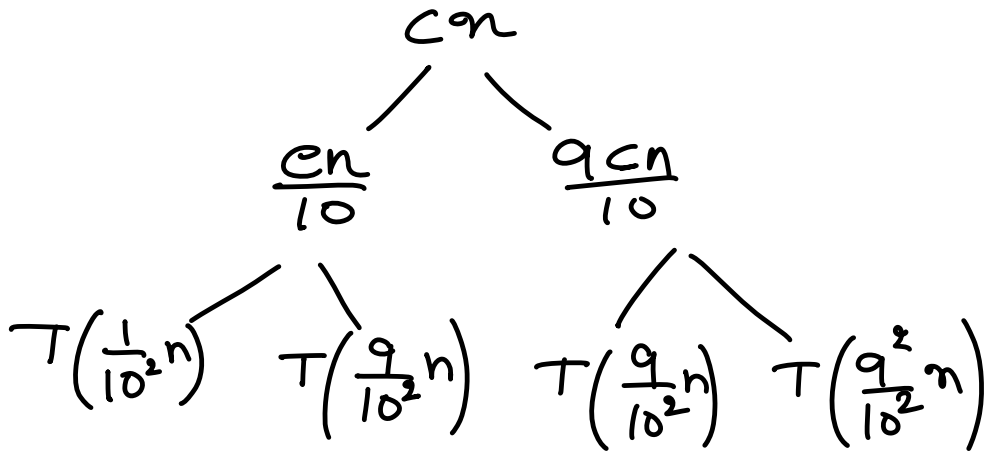
$$= T\left(\frac{1}{10}n\right) + T\left(\frac{9}{10}n\right) + cn.$$

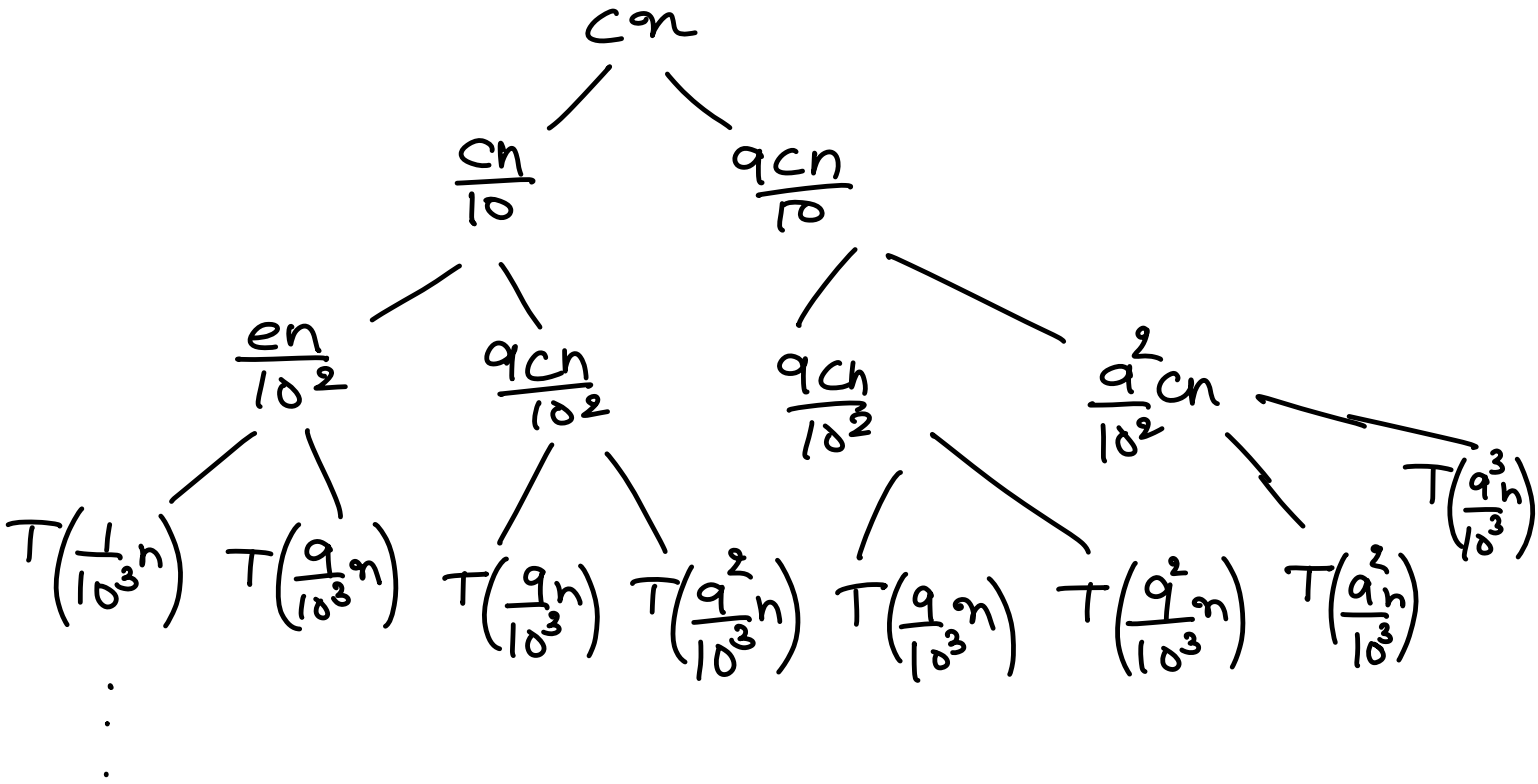
Let $\Theta(n) = cn$.

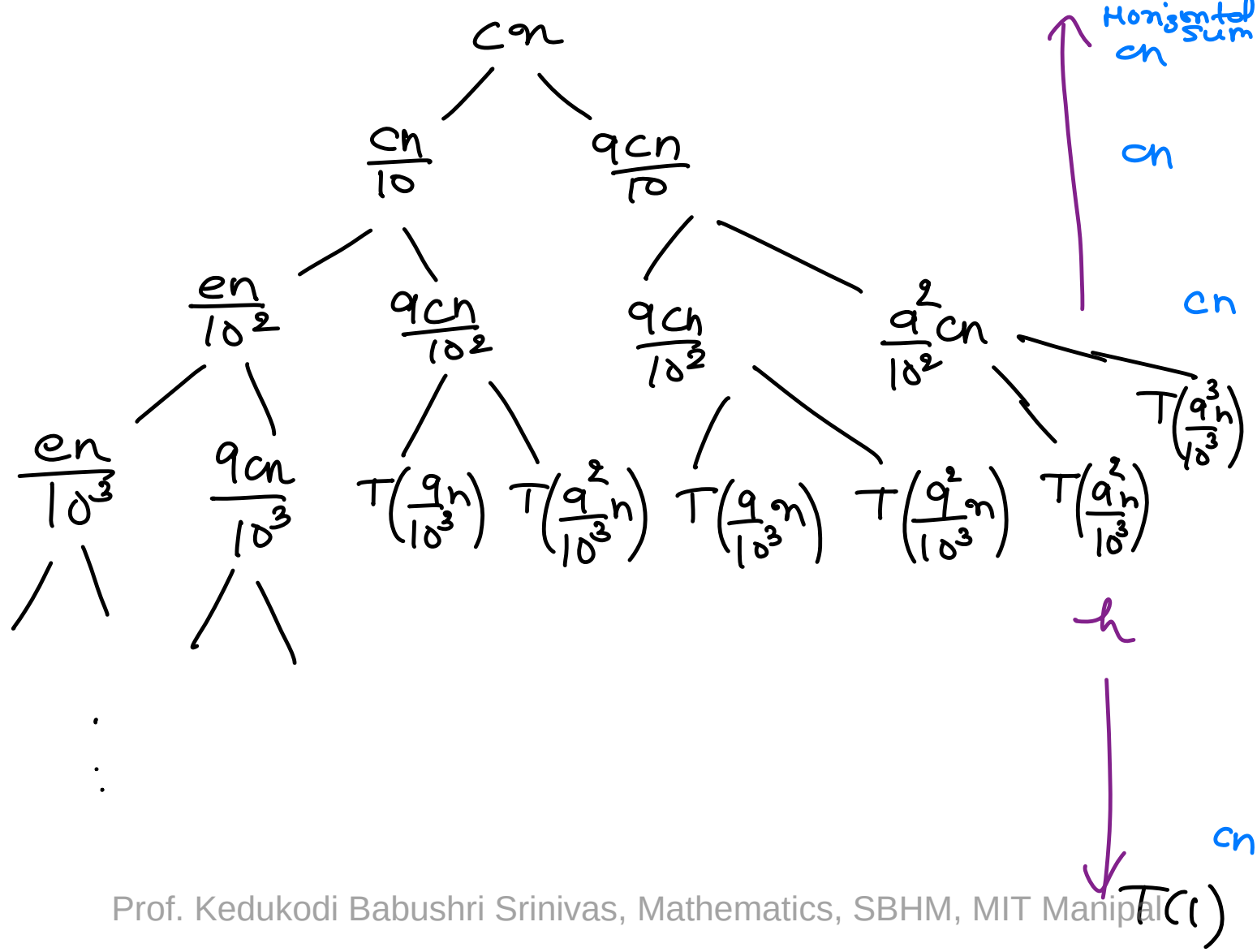
$$T\left(\frac{1}{10}n\right) = T\left(\frac{1}{10^2}n\right) + T\left(\frac{9}{10^2}n\right) + \frac{cn}{10}$$

$$T\left(\frac{9}{10}n\right) = T\left(\frac{9}{10^2}n\right) + T\left(\frac{9^2}{10^2}n\right) + \frac{9cn}{10}$$


cn
$T\left(\frac{1}{10}n\right)$
$T\left(\frac{9}{10}n\right)$
$T\left(\frac{1}{10^2}n\right)$
$T\left(\frac{9}{10^2}n\right)$
$T\left(\frac{9}{10^2}n\right)$
$T\left(\frac{9^2}{10^2}n\right)$
$T\left(\frac{9^3}{10^3}n\right)$
$T\left(\frac{1}{10^3}n\right)$
$T\left(\frac{9}{10^3}n\right)$
$T\left(\frac{9}{10^3}n\right)$
$T\left(\frac{9^2}{10^3}n\right)$
$T\left(\frac{9}{10^3}n\right)$
$T\left(\frac{9^2}{10^3}n\right)$
$T\left(\frac{9^2}{10^3}n\right)$


cn
cn/10
9cn/10
T(1/10² n)
T(9/10² n)
T(9/10² n)
T(9²/10² n)


cn
cn/10
9cn/10
en/10^2
9cn/10^2
9cn/10^2
a^2cn/10^2
T(1/10^3 n)
T(9/10^3 n)
T(9/10^3 n)
T(a^2/10^3 n)
T(9/10^3 n)
T(a^2/10^3 n)
T(a^2/10^3 n)
T(a^3/10^3 n)


cn
cn/10
9cn/10
cn/10^2
9cn/10^2
9cn/10^2
a^2cn/10^2
cn/10^3
9cn/10^3
T(9n/10^3)
T(9^2n/10^3)
T(9n/10^3)
T(9^2n/10^3)
T(9^2n/10^3)
T(9^3n/10^3)
Horizontal sum
cn
cn
cn
h
cn
T(1)

Let height of the tree = h.

Here $T(n)$ = Sum of all nodes of the tree.

$$= \underbrace{cn + cn + \cdots + cn}_{h \text{ times}}$$

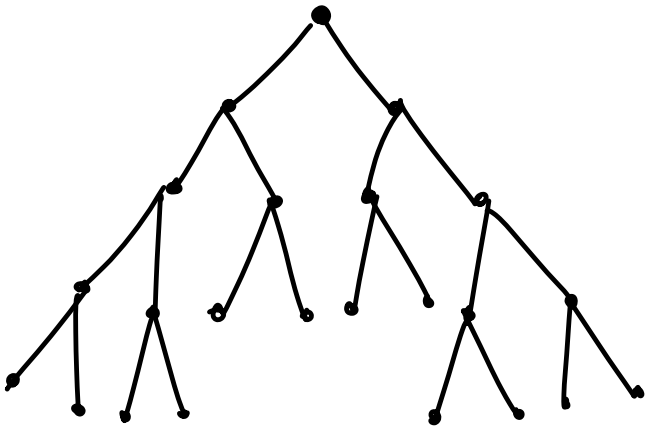
Note: The height on left side will be smaller than the right side.

Left: $\lfloor \log_{10} n \rfloor$ $\quad h$

Right: $\lfloor \log_{10/9} n \rfloor$ $\quad H$

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Height of a tree



		Maximum No of Leaves
$h=0$	level 1	0
$h=1$		2
$h=2$		4
$h=3$		8
$h=4$		16

$T(n) =$ Total of all the nodes

$= (cn)\, H$

$= (cn)\left\lfloor \log_{\frac{10}{9}} n \right\rfloor \in O(n \log n).$

Average-Case Analysis :

partitioning time

$$T(n) = \begin{cases} T(0) + T(n-1) + \Theta(n), & 0 : n-1 \text{ split} \\ T(1) + T(n-2) + \Theta(n), & 1 : n-2 \\ \vdots & \\ T(k) + T(n-k-1) + \Theta(n), & k : n-k-1 \\ \vdots & \\ T(n-1) + T(0) + \Theta(n), & n-1 : 0 \text{ split} \end{cases}$$

n cases

$k = 0, 1, \ldots, n-1.$

$k + n - k - 1 = n-1$

$E[T(n)] = ?$ pdf, $p(x) = \frac{1}{n}$

$T(n)$ can be written as

$$T(n) = \sum_{k=0}^{n-1} X_k \left(T(k) + T(n-k-1) + \Theta(n) \right)$$

X_k – indicator random variable for k^{th} split
(indicates whether k^{th} split occured or not)

ie $X_k = \begin{cases} 1 & \text{if } k : n-k-1 \text{ split occurs} \\ 0 & \text{otherwise.} \end{cases}$

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$$E[T(n)] = E\left[\sum_{k=0}^{n-1} X_k \left(T(k) + T(n-k-1) + \Theta(n)\right)\right]$$

$$= \sum_{k=0}^{n-1} E\left[X_k \left(T(k) + T(n-k-1) + \Theta(n)\right)\right]$$

$$= \sum_{k=0}^{n-1} E[X_k]\, E\left[T(k) + T(n-k-1) + \Theta(n)\right]$$

$$= \sum_{k=0}^{n-1} E[X_k] \left(E[T(k)] + E[T(n-k-1)] + E[\Theta(n)]\right)$$

$$X_k = \begin{cases} 1 & k : n-k-1 \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

x

$$E[X_k] = \sum_{x \in X} x\, p(x)$$

$$= 1 \cdot P(k : n-k-1 \text{ split occurs}) + 0 \cdot P(k : n-k-1 \text{ split does not occur})$$

$$= 1 \cdot \left(\frac{1}{n}\right) + 0 \cdot \left(1 - \frac{1}{n}\right)$$

uniform distribution

$\therefore \sum p(x) = 1$

$$E[T(n)] = \sum_{k=0}^{n-1} \frac{1}{n} \left(E[T(k)] + E[T(n-k-1)] + \Theta(n) \right)$$

$E[\text{constant}] = \text{constant}$.

$$= \sum_{k=0}^{n-1} \frac{1}{n} \left[E[T(k)] + E[T(n-k-1)] \right] + \sum_{k=0}^{n-1} \frac{1}{n} \Theta(n).$$

$$= \sum_{k=0}^{n-1} \frac{2}{n} E[T(k)] + \cancel{n} \cdot \frac{\Theta(n)}{\cancel{n}}$$

$$= \frac{2}{n} \sum_{k=0}^{n-1} E[T(k)] + \Theta(n) \quad \text{---} \quad (*)$$

why? ↑

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Nok:

$$\sum_{k=0}^{n-1} E[T(k)] + E[T(n-k-1)]$$

✓ Reason

$$= E[T(0)] + E[T(n-1)]$$

$$+ E[T(1)] + E[T(n-2)]$$

$$+ \ldots$$

$$+ \underbrace{E[T(n-1)]} + \underbrace{E[T(0)]}$$

$$= \sum_{k=0}^{n-1} E(T(k)) + \sum_{k=0}^{n-1} E(T(k)) = 2\sum_{k=0}^{n-1} E[T(k)]$$

Using induction to compute $E[T(n)]$.

Good Guess:

Using the recursion trees (eg 10:90 case)

the guess is $E(T(n)) \in O(n \log n)$.

Assume $E[T(n)] = O(n \log n)$

Test the result holds for small values of n.

Induction hypothesis:

Assume that it holds for all values less than n.

To prove : the result holds for n.

$$E[T(n)] = \frac{2}{n} \sum_{k=0}^{n-1} E[T(k)] + \theta(n)$$

$$= \frac{2}{n} \sum_{k=2}^{n-1} k \log k + \theta(n)$$